Camera: the optical system

Topics

- **Introduction and the Camera Optical System**
- Planar (2D) Projective Geometry
- Spatial (3D) Projective Geometry
- Camera Geometry (3D → 2D Projection)

OPTICAL SENSOR: CAMERA

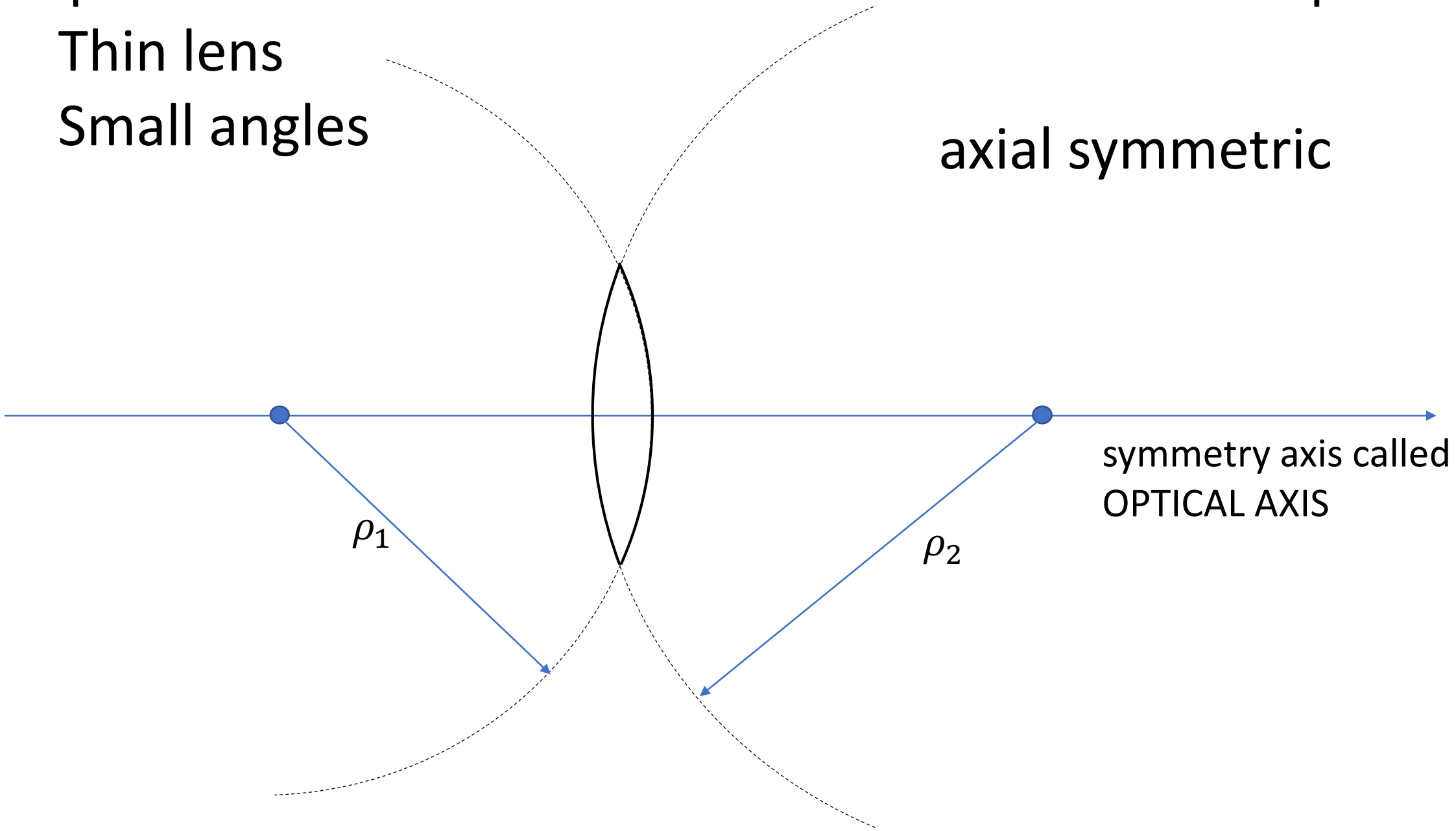
- Screen with (order of 10^6 or more) photosensitive elements called *pixels* (PIXEL = PICture ELement)
  - Optical → electric transducers
- **Optical system** enables the selection of a single direction of incoming light at each element
- Electric circuits that collect the signal generated at the pixels

- 30-60 frames per second (even more in special cameras)

simplified camera model: lens = intersection of two spheres

- Thin lens
- Small angles

axial symmetric

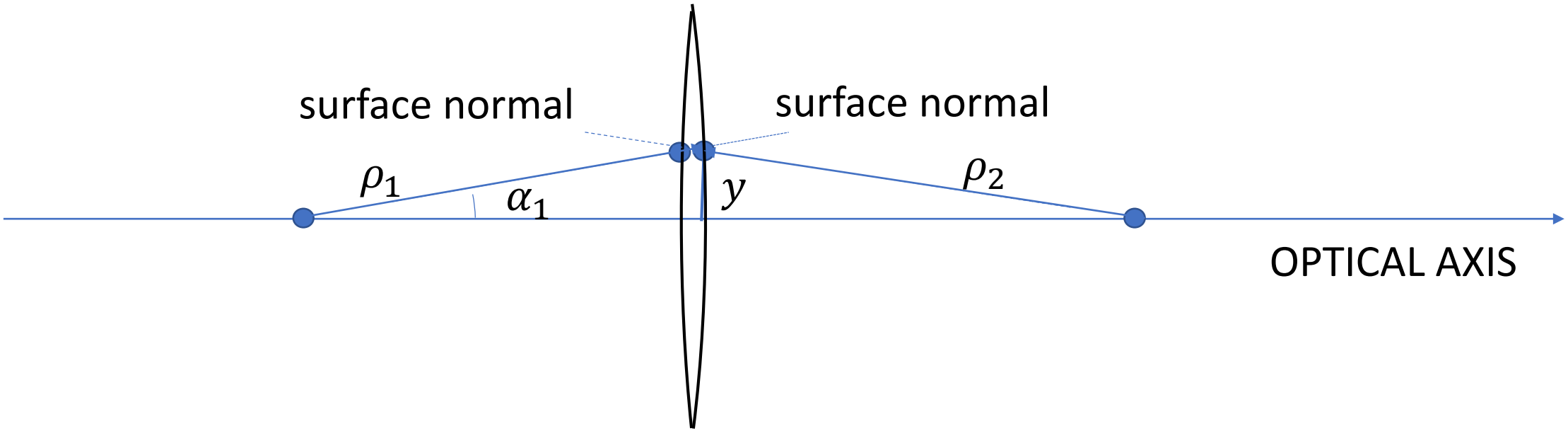


simplified camera model: lens = intersection of two spheres

- **Thin lens**
- **Small angles**

Hp: small angles → $\alpha_1 = y_1/\rho_1$

and also: $\alpha_2 = -y_2/\rho_2$

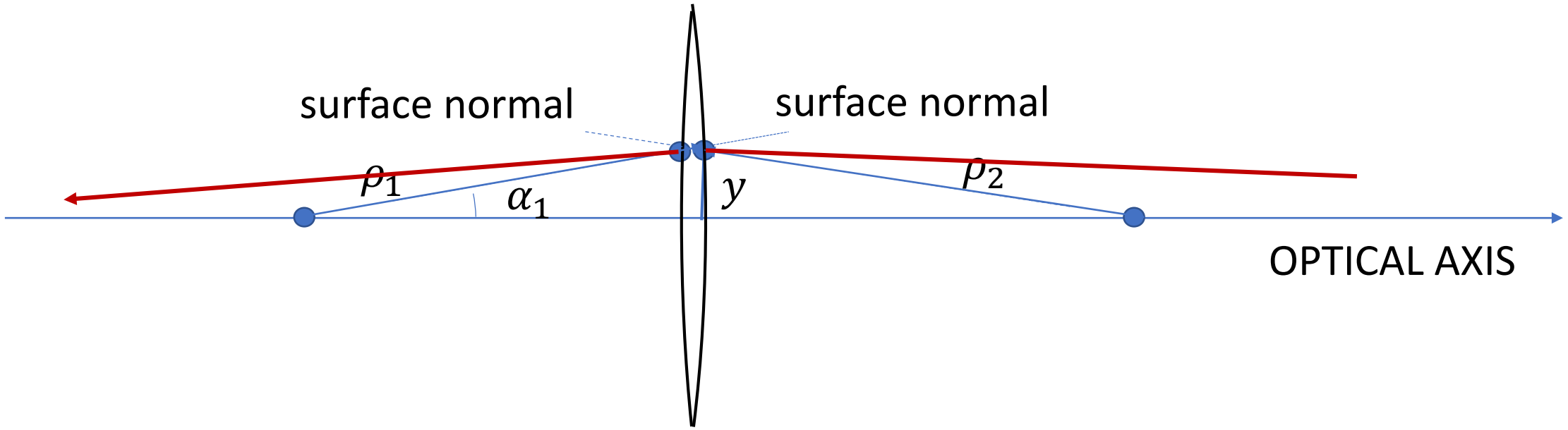


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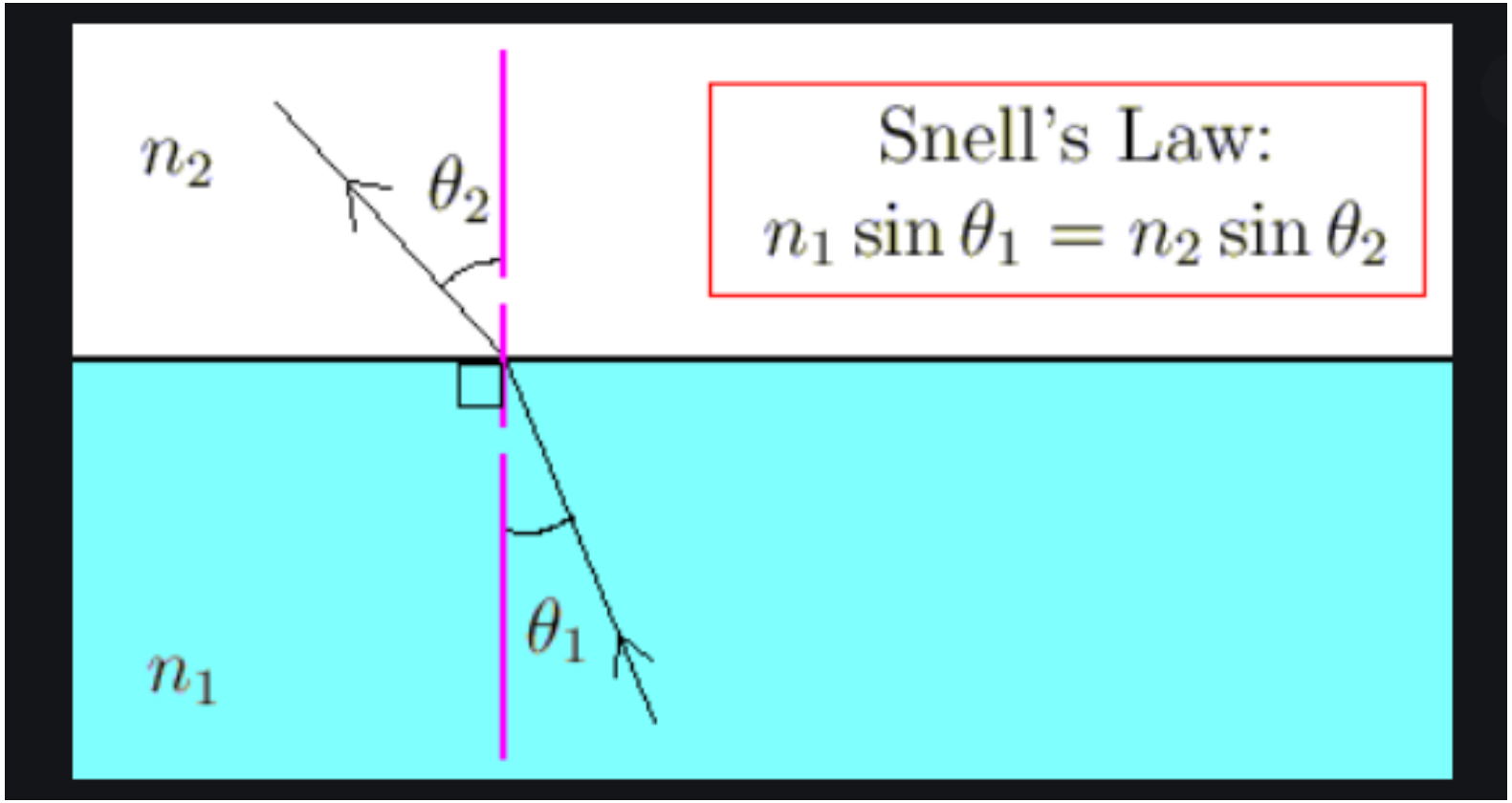
Light ray: 1. air-glass refraction at y_1 2. glass-air refraction at y_2

Hp: thin lens → very short light path within the lens → $y_1 = y_2 = y$

The deviation of a light ray crossing the lens

Refraction

A light ray crosses the boundary between two media

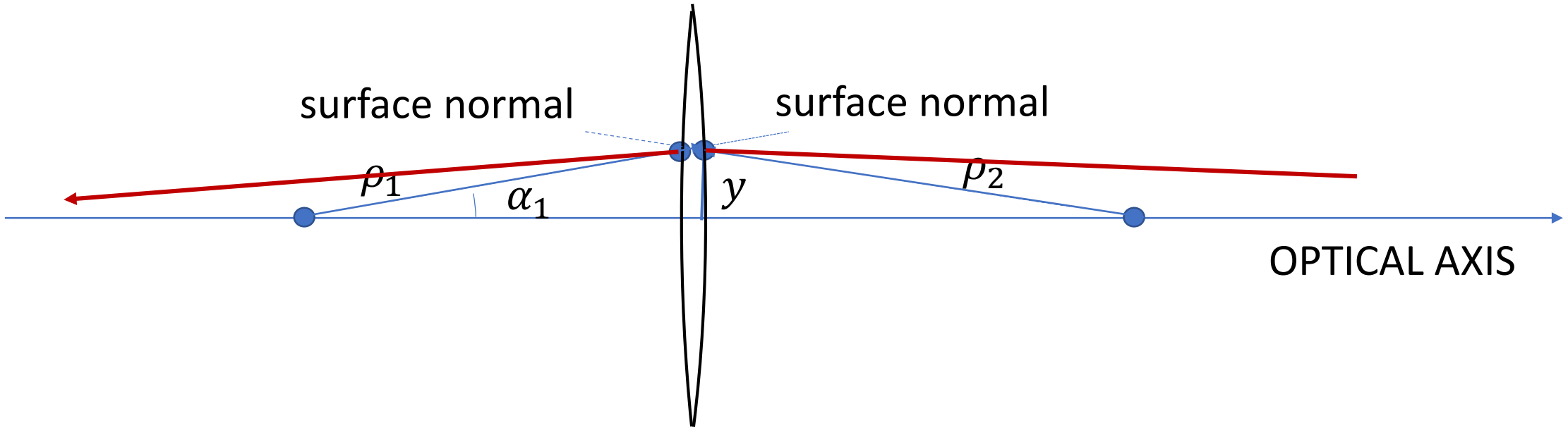


simplified camera model: lens = intersection of two spheres

- **Thin lens**
- **Small angles**

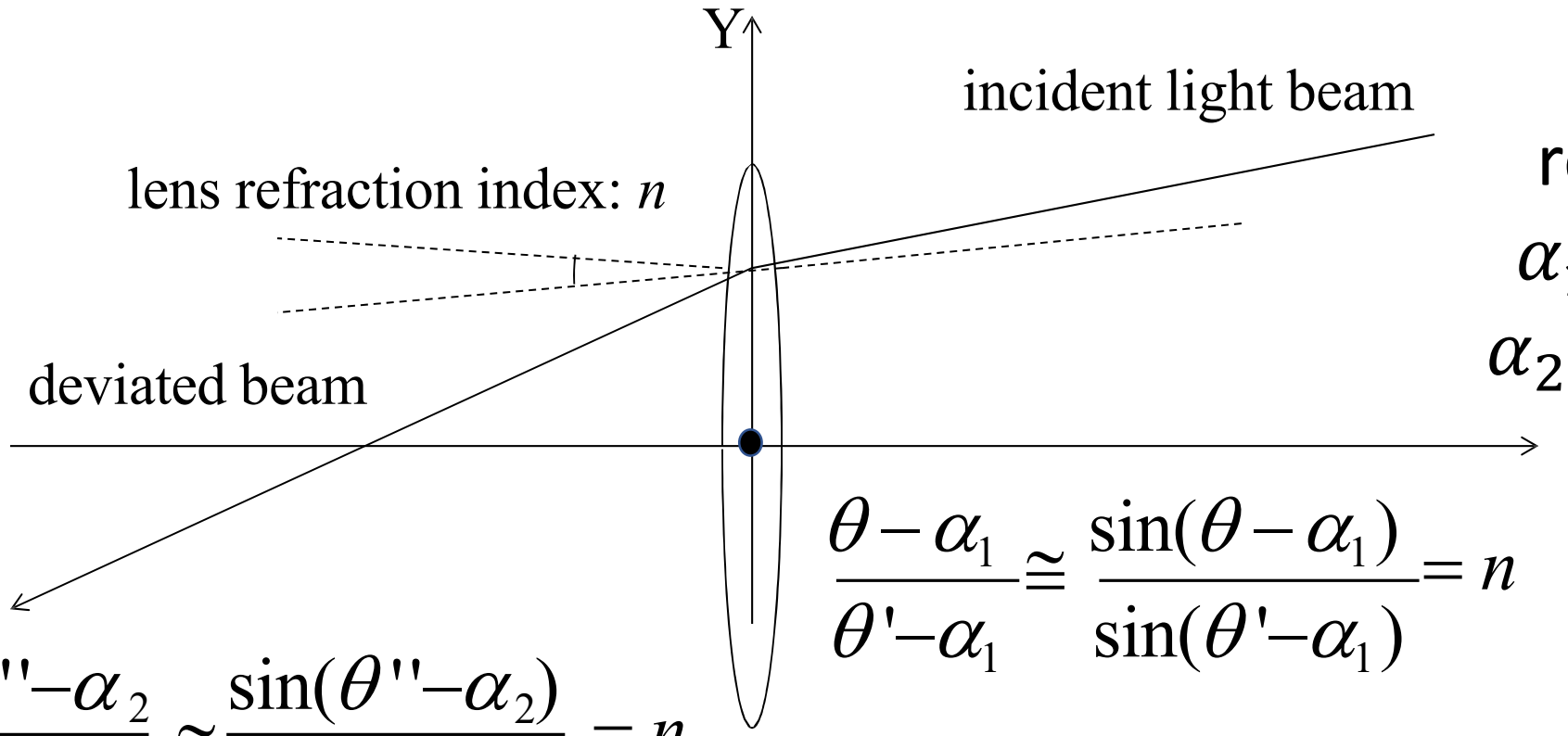
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Light ray: 1. air-glass refraction at y_1 2. glass-air refraction at y_2

Hp: thin lens → very short light path within the lens → $y_1 = y_2 = y$



remember

$\alpha_1 = y_1/\rho_1$

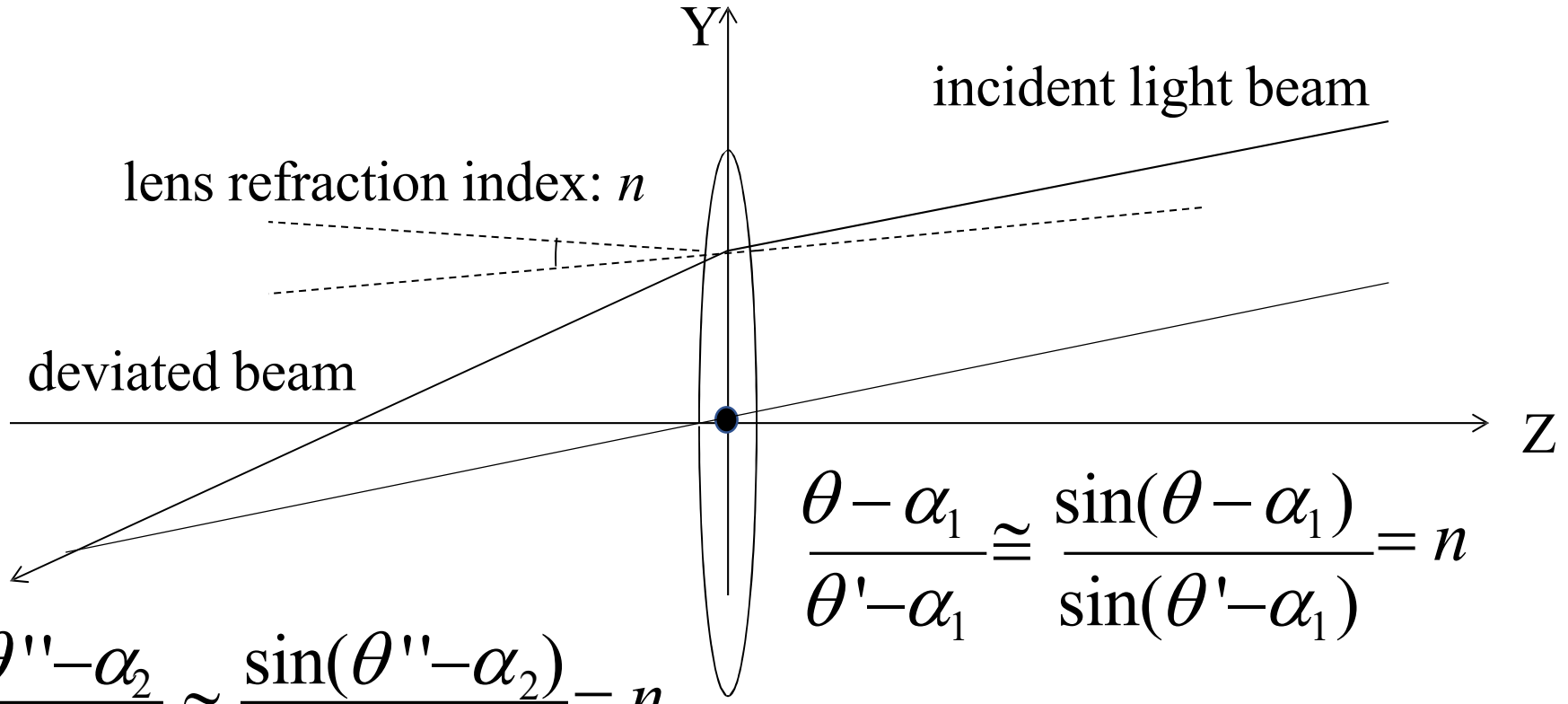
$\alpha_2 = -y_2/\rho_2$

$$\frac{\theta-\alpha_1}{\theta'-\alpha_1} \cong \frac{\sin(\theta-\alpha_1)}{\sin(\theta'-\alpha_1)} = n$$

$$\frac{\theta''-\alpha_2}{\theta'-\alpha_2} \cong \frac{\sin(\theta''-\alpha_2)}{\sin(\theta'-\alpha_2)} = n$$

deviation angle ? $\delta\,\theta = \theta'' - \theta$

$$\delta\,\theta = (n-1)Y\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)$$



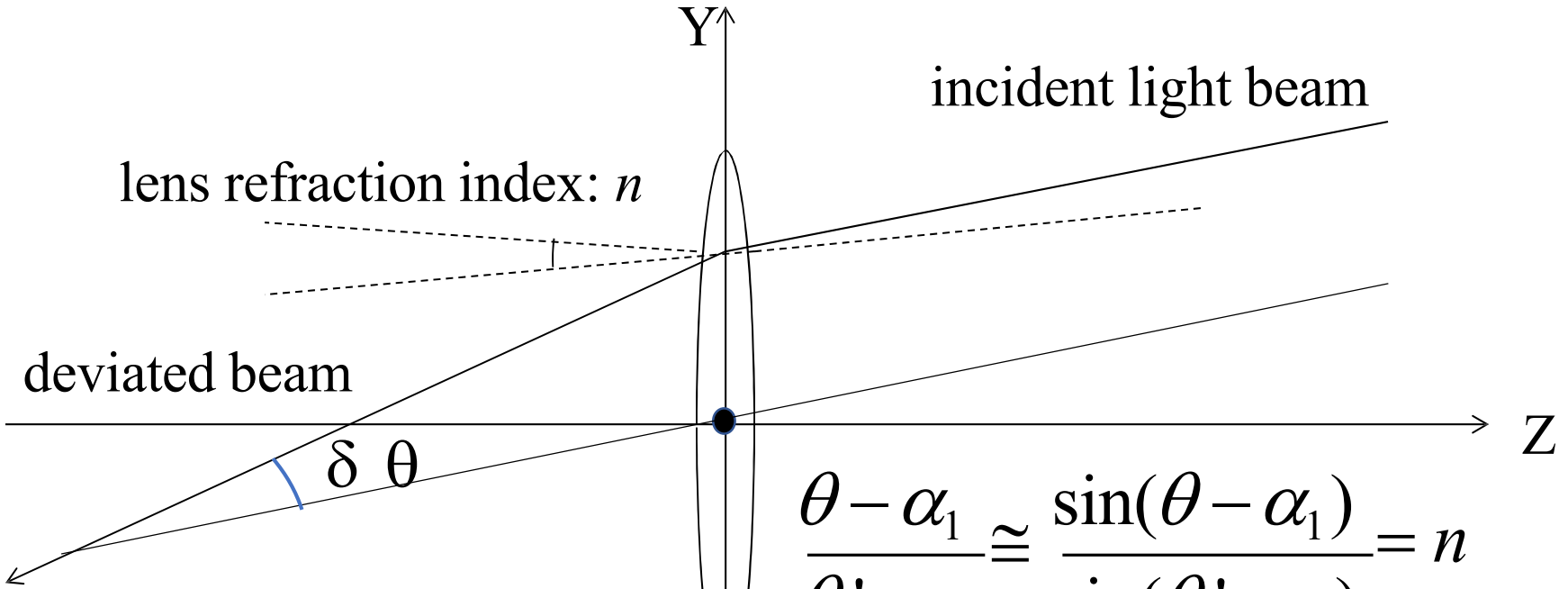
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deviation angle ? $\delta\,\theta = \theta'' - \theta$

$$\delta\,\theta = (n-1)Y\left(\frac{1}{\rho_1}+\frac{1}{\rho_2}\right)$$

$Y = 0 \rightarrow \delta\,\theta = 0$ undeviated ray



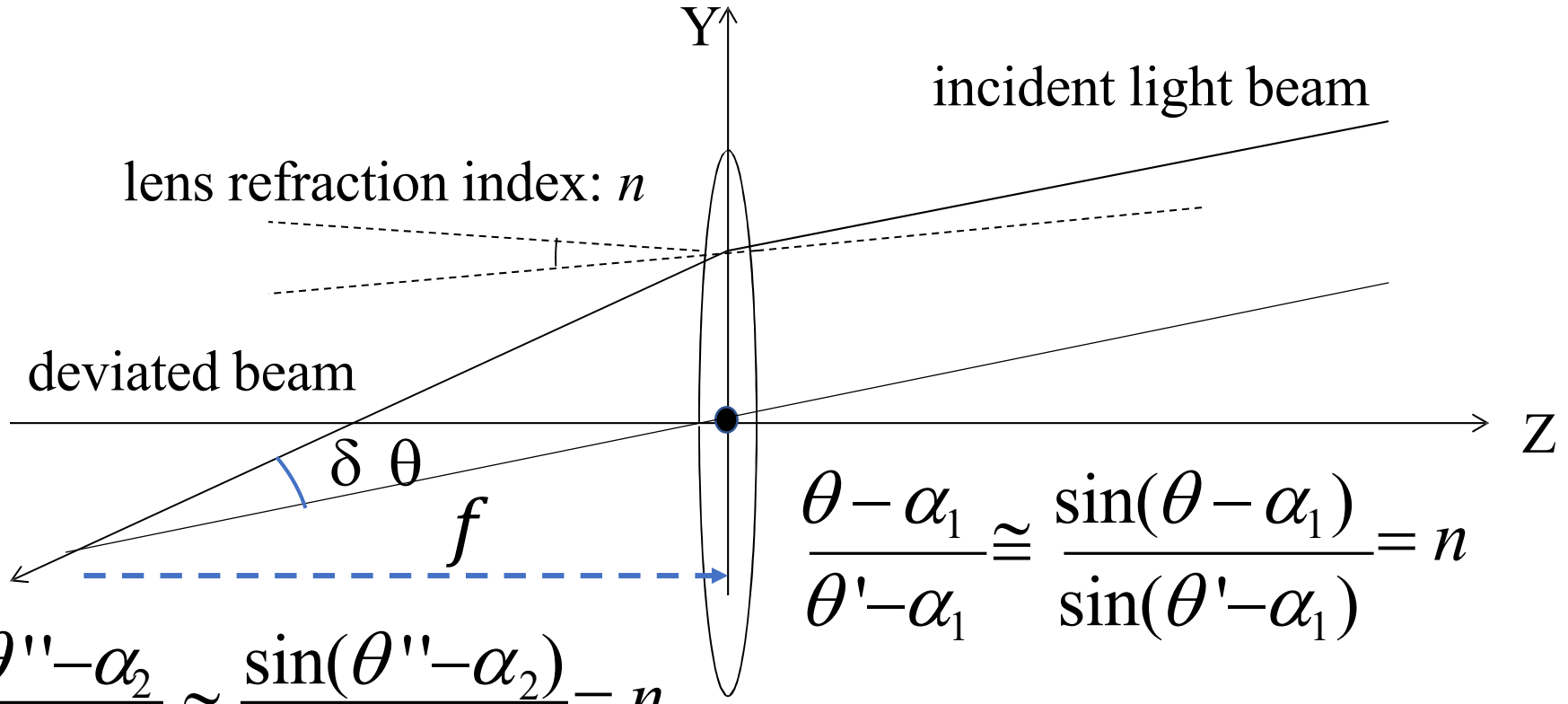
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deviation angle ? $\delta\ \theta\ = \theta'' - \theta$

$$\delta\ \theta = (n\ -\ 1\ )Y\ (\frac{1}{\rho_1} + \frac{1}{\rho_2})$$

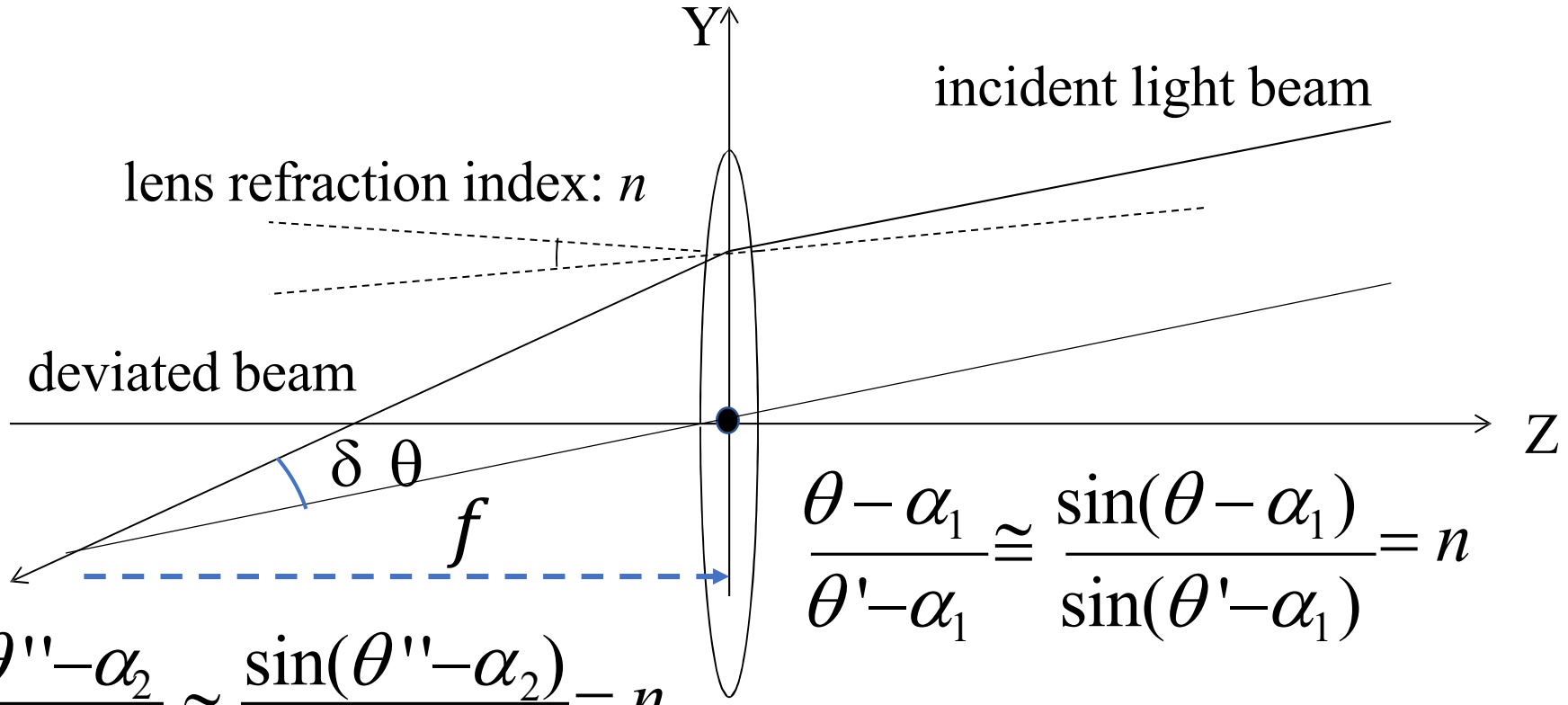
$Y = 0 \rightarrow \delta\ \theta = 0$ undeviated ray



deviation angle ? $\delta\ \theta\ = \theta'' - \theta$

$$\delta\ \theta = (n - 1)Y\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)$$

$Y = 0 \rightarrow \delta\ \theta = 0$ undeviated ray



$$\frac{\theta-\alpha_1}{\theta'-\alpha_1} \cong \frac{\sin(\theta-\alpha_1)}{\sin(\theta'-\alpha_1)} = n$$

$$\frac{\theta''-\alpha_2}{\theta'-\alpha_2} \cong \frac{\sin(\theta''-\alpha_2)}{\sin(\theta'-\alpha_2)} = n$$

deviation angle ? $\delta\ \theta\ = \theta'' - \theta$

$$\delta\ \theta = (n - 1)Y\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)$$

$$f\ \ \delta\ \theta = Y$$

Focalization (convergence) of parallel rays for small angles

$$f \ \delta\theta = \ Y$$

with

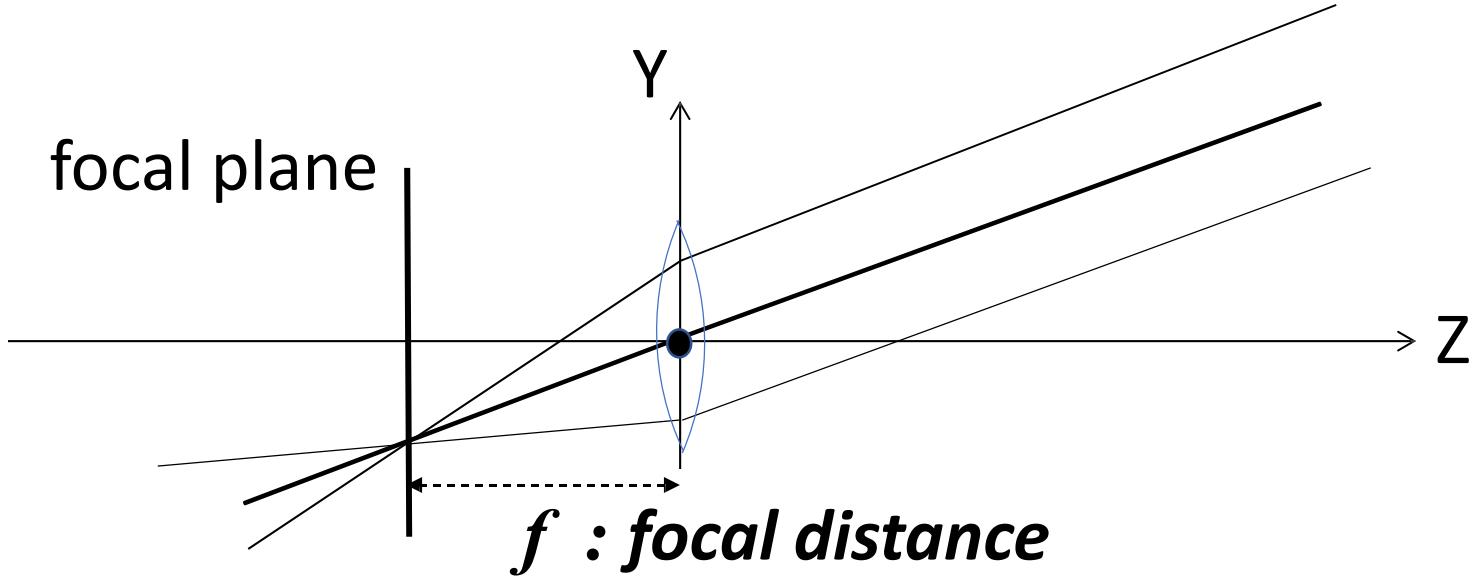
$$f = \frac{1}{(n - 1)\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)}$$

focal distance

Focalization of parallel rays

$$f = \frac{1}{(n - 1)\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)}$$

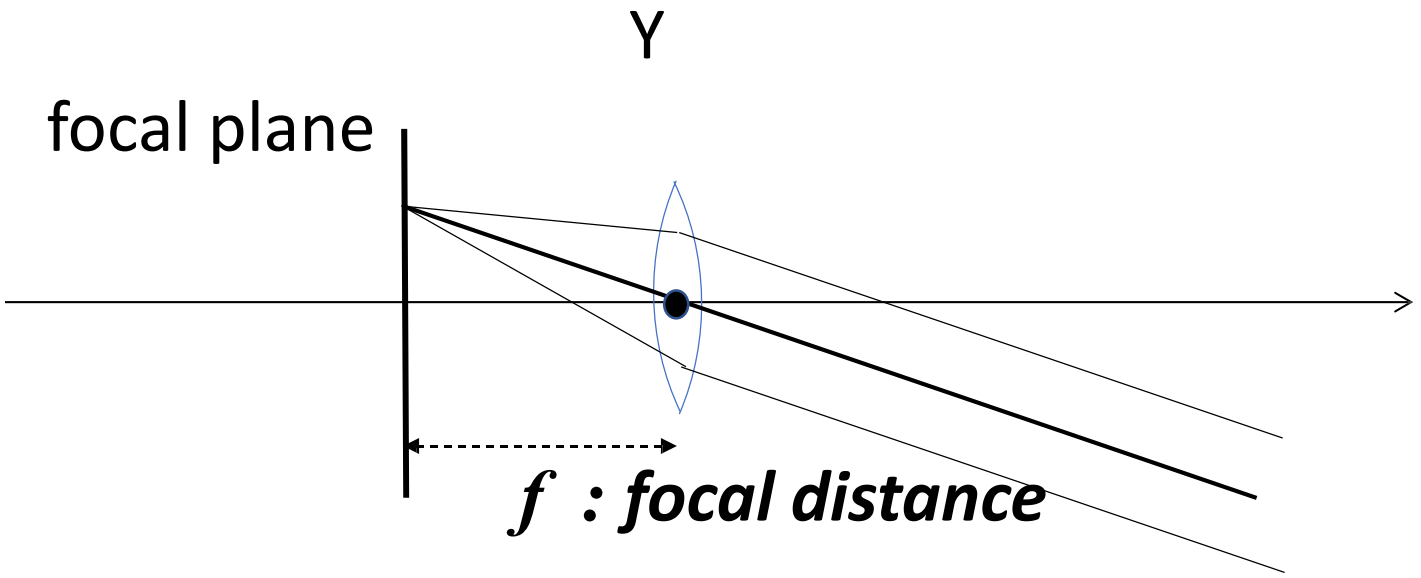
ALL PARALLEL RAYS CROSSING THE LENS concur at a common point on the a *focal* plane $Z = -f$



Focalization of parallel rays: same plane for any direction

$$f = \frac{1}{(n - 1)\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)}$$

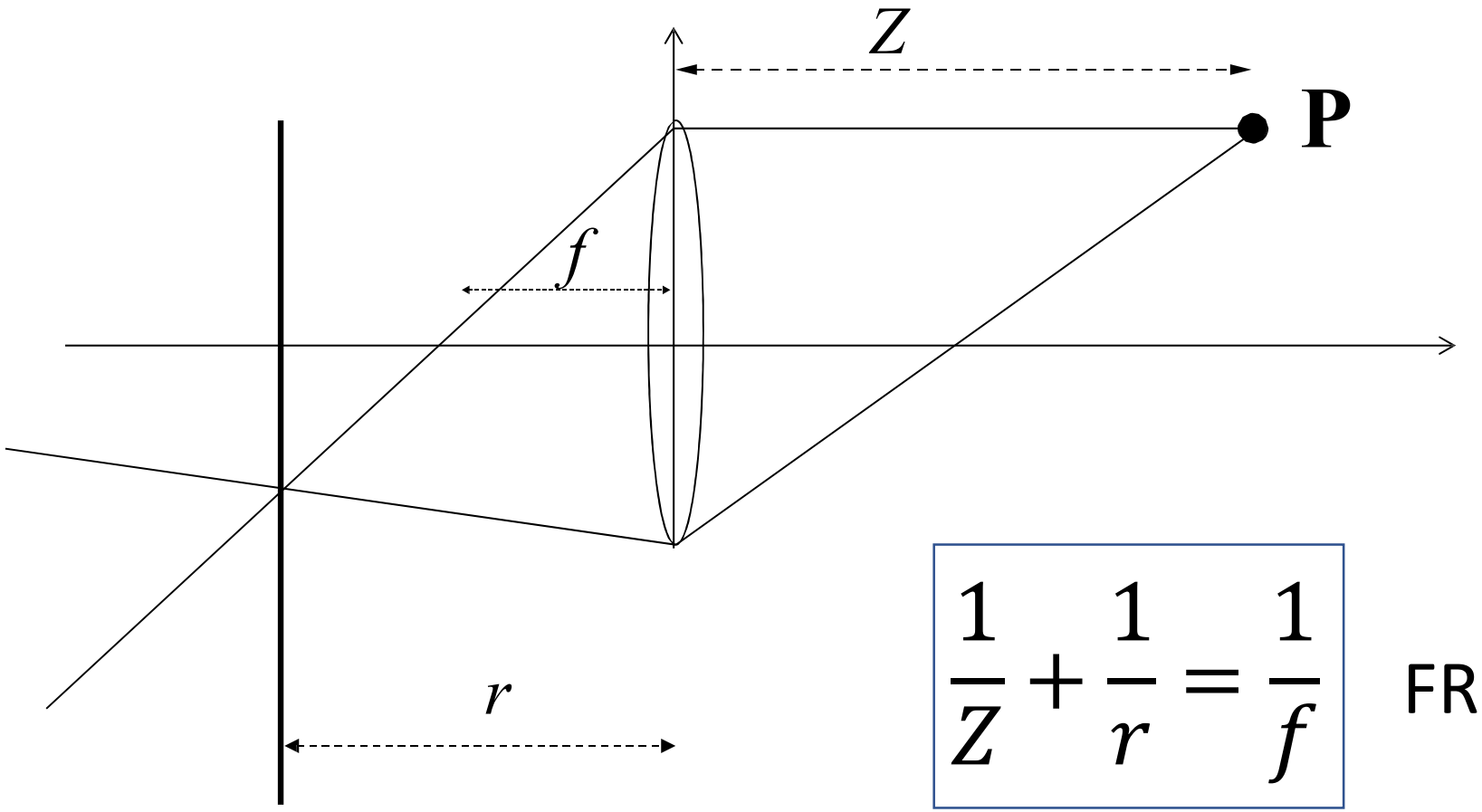
ALL PARALLEL RAYS CROSSING THE LENS concur at a common point on the a *focal* plane $Z = -f$



Parallel rays = rays coming from a single point at ∞ distance

→ focalized at a point on plane $Z = -f$

Focalization of rays coming from a single point at **finite** distance



$$\frac{1}{Z} + \frac{1}{r} = \frac{1}{f}$$

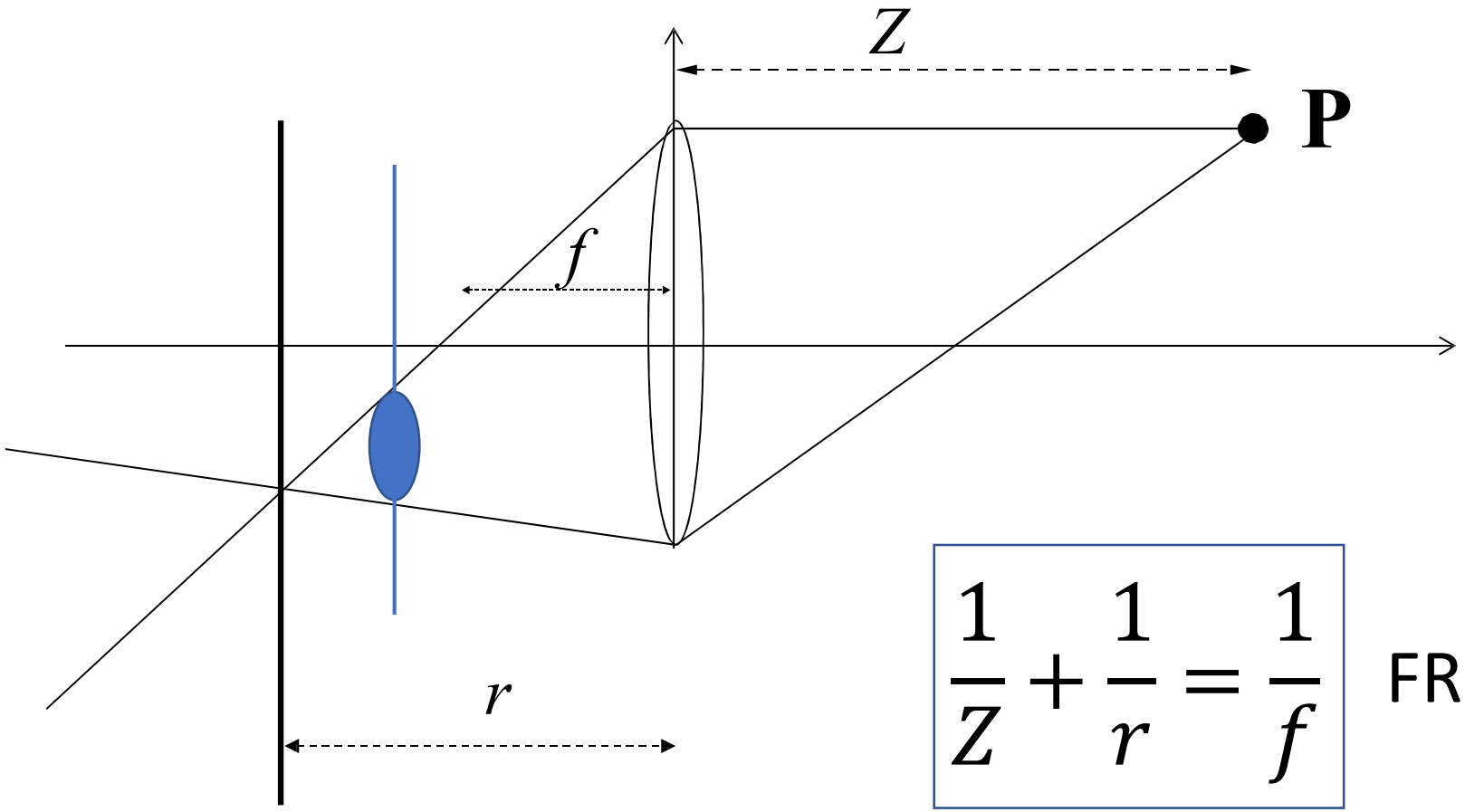
FRESNEL LAW

Where sould the screen be placed ?

Where sould the screen be placed

diffusion of rays coming from a single point at finite distance

if the screen is NOT placed at $Z = r$ → blurred image



$$\frac{1}{Z} + \frac{1}{r} = \frac{1}{f}$$

FRESNEL LAW

PIN-HOLE camera model:

- Thin lens
- Small angles

+

Assume that

- Observed point are *distant* from the lens, namely

$$\boldsymbol{Z(P) >> \text{lens aperture}}$$

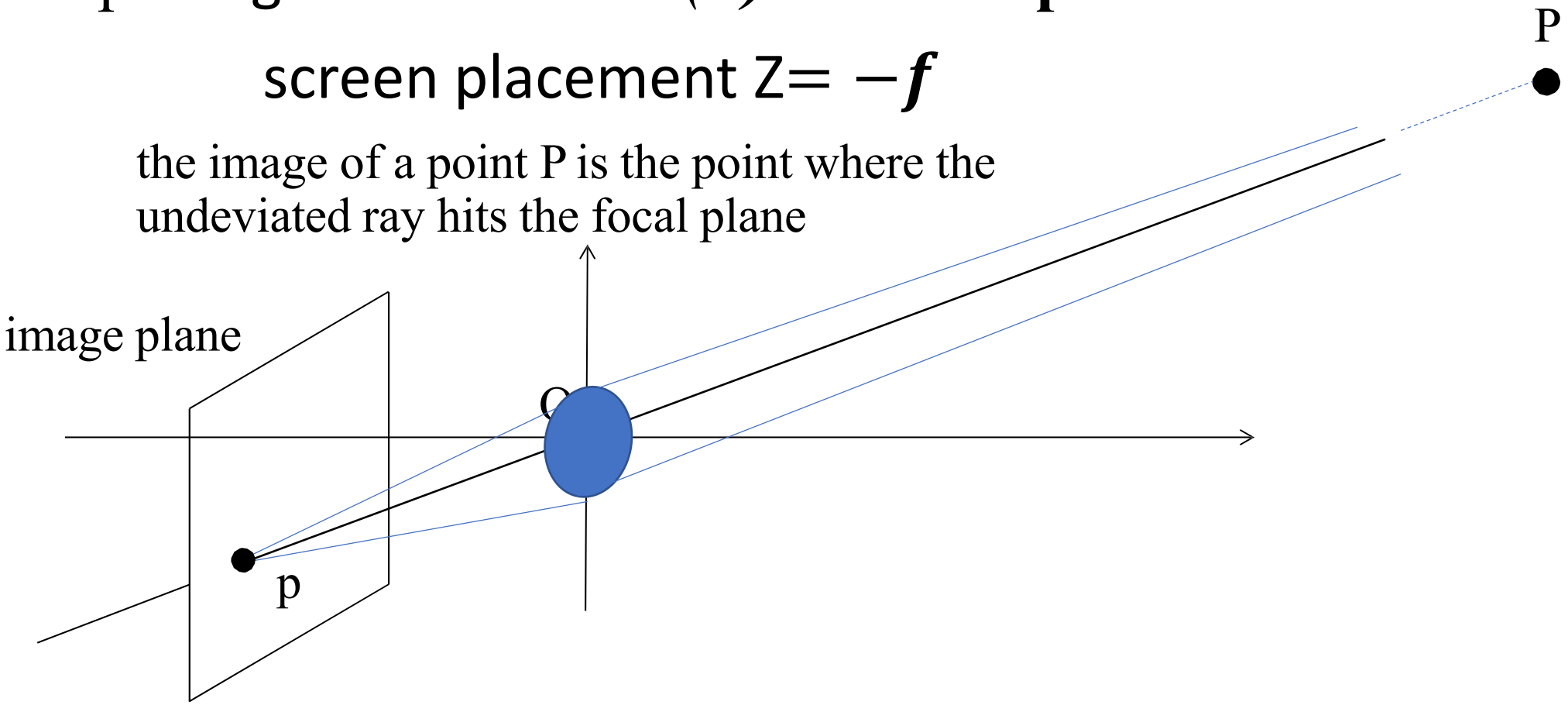
→

Place the screen where **parallel** rays converge,

i.e. at $\boldsymbol{Z = -f}$

Hp: «large» distances $Z(P) >>$ lens aperture

screen placement $Z = -f$



p = image of P is on the line(P,O) and on the image plane

p = image of P = image plane ∩ line(O,P)

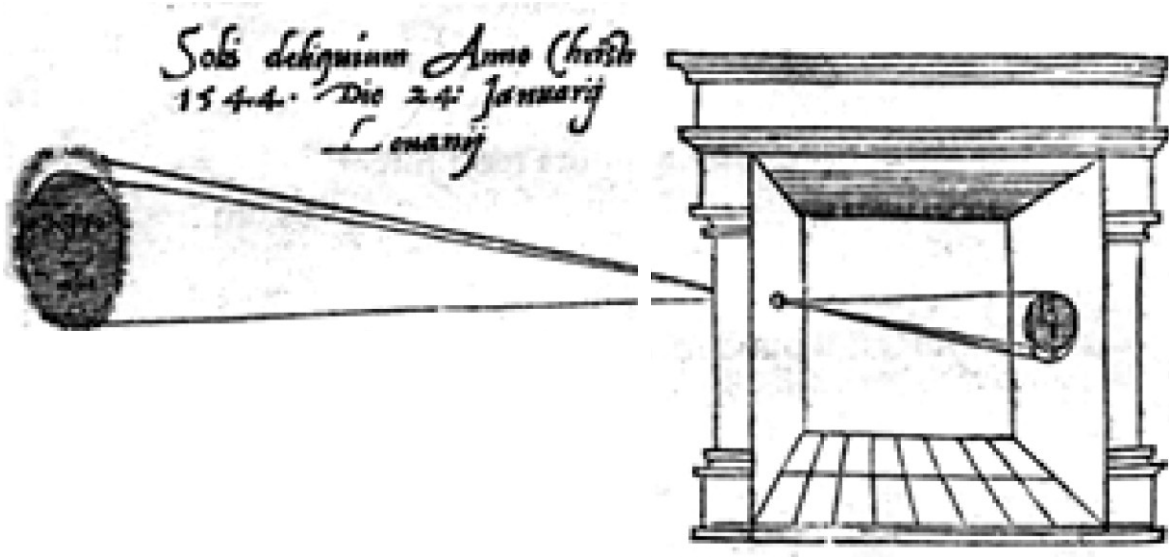
Model: the pin-hole camera model

Our simplified model:

(i) thin spherical lens, (ii) *small angles*, (iii) $Z(P) >> a$, (iv) $Z = f$

→ PIN-HOLE CAMERA

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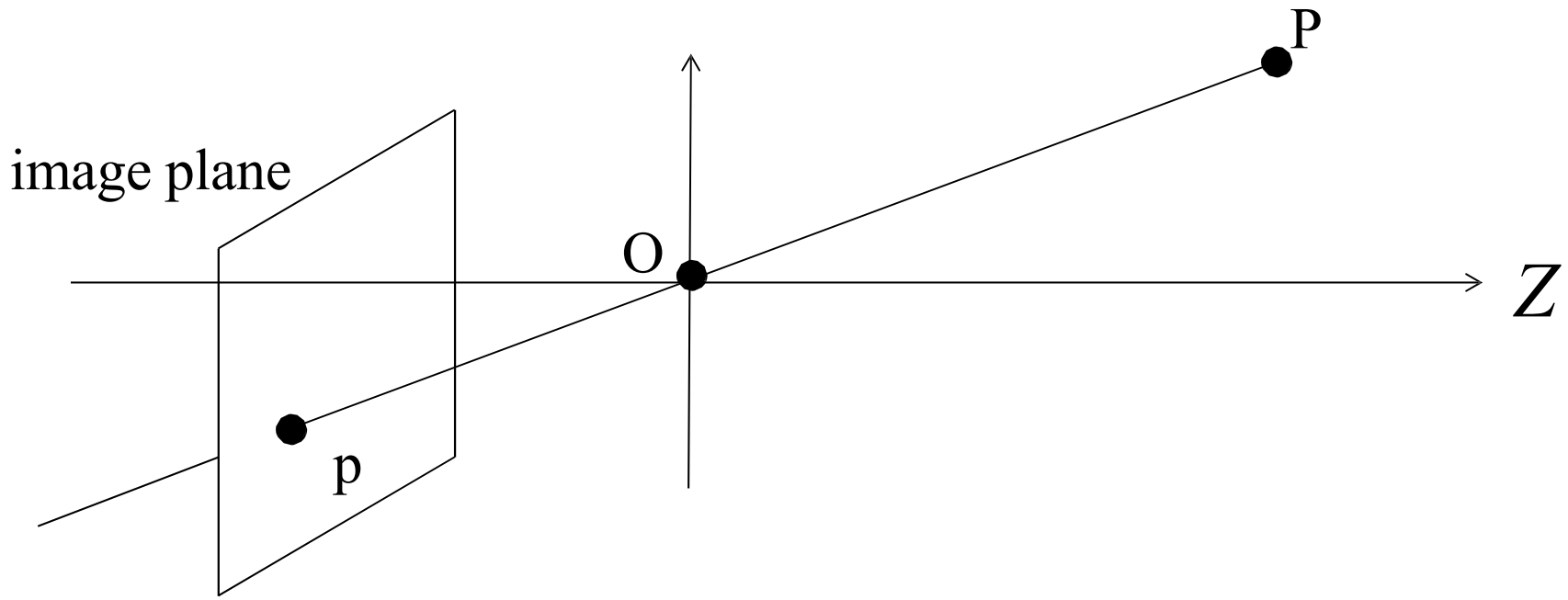
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The image of a point P in the scene

Hp: «large» distances ***Z(P)*** **>> lens aperture**

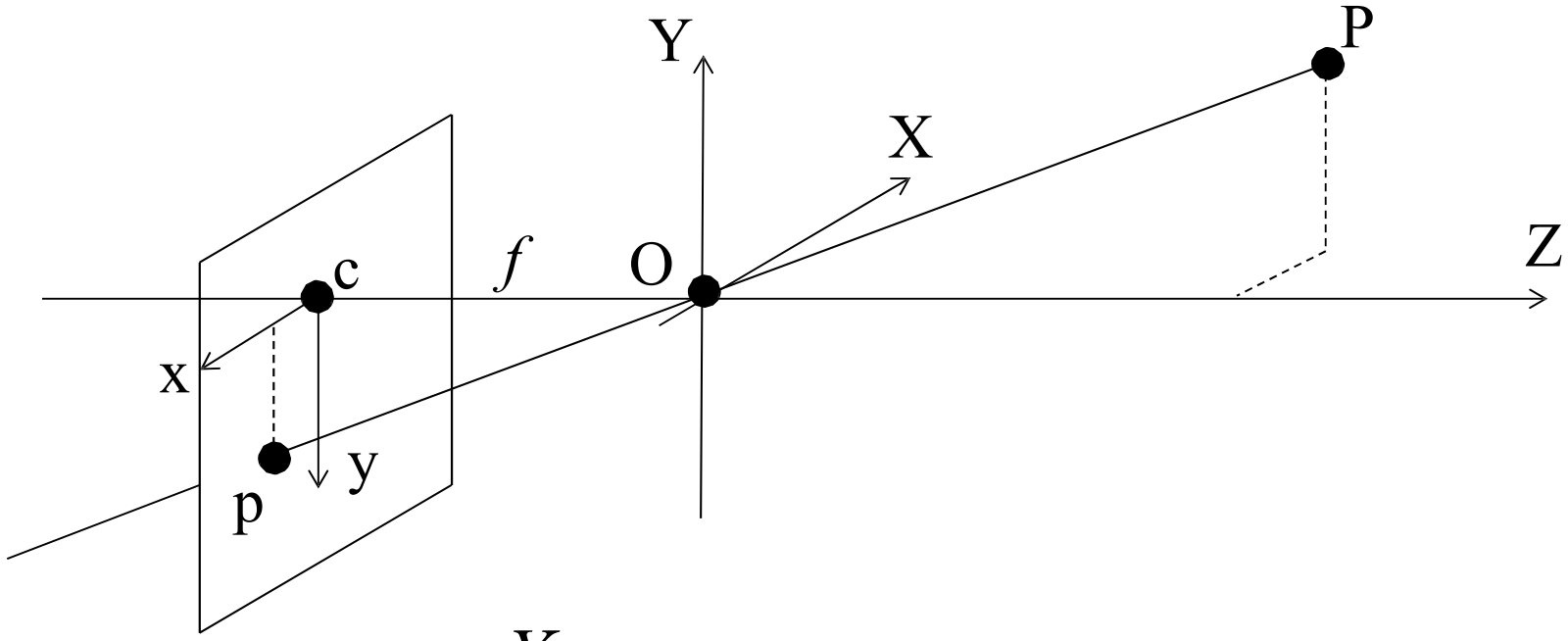
screen placement Z= $-f$

the image of a point P belongs to the line (P,O)



p = image of P = is on the image plane and on the line(O,P) → p = image plane ∩ line(O,P)

viewing ray of p: line(O,p) =
locus of the scene points projecting onto image point p



$$x = f\,\frac{X}{Z}$$

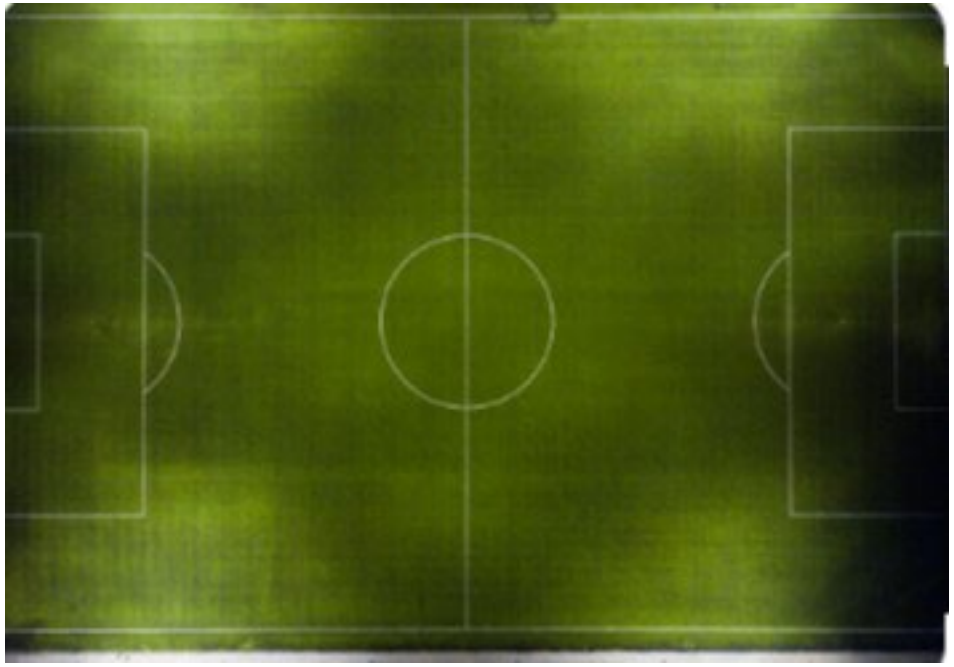
$$y = f\,\frac{Y}{Z}$$

central projection

-nonlinear
-not shape-preserving
-not length-ratio preserving

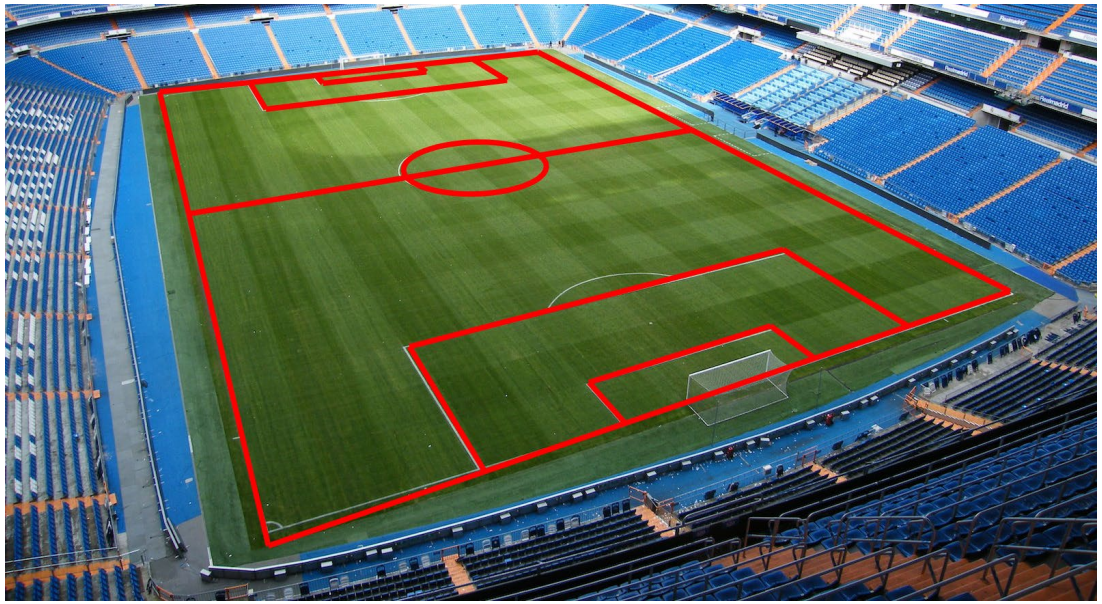
Scenario that we imagine when we think at the image of something:

image = «a scaled version of the observed object»

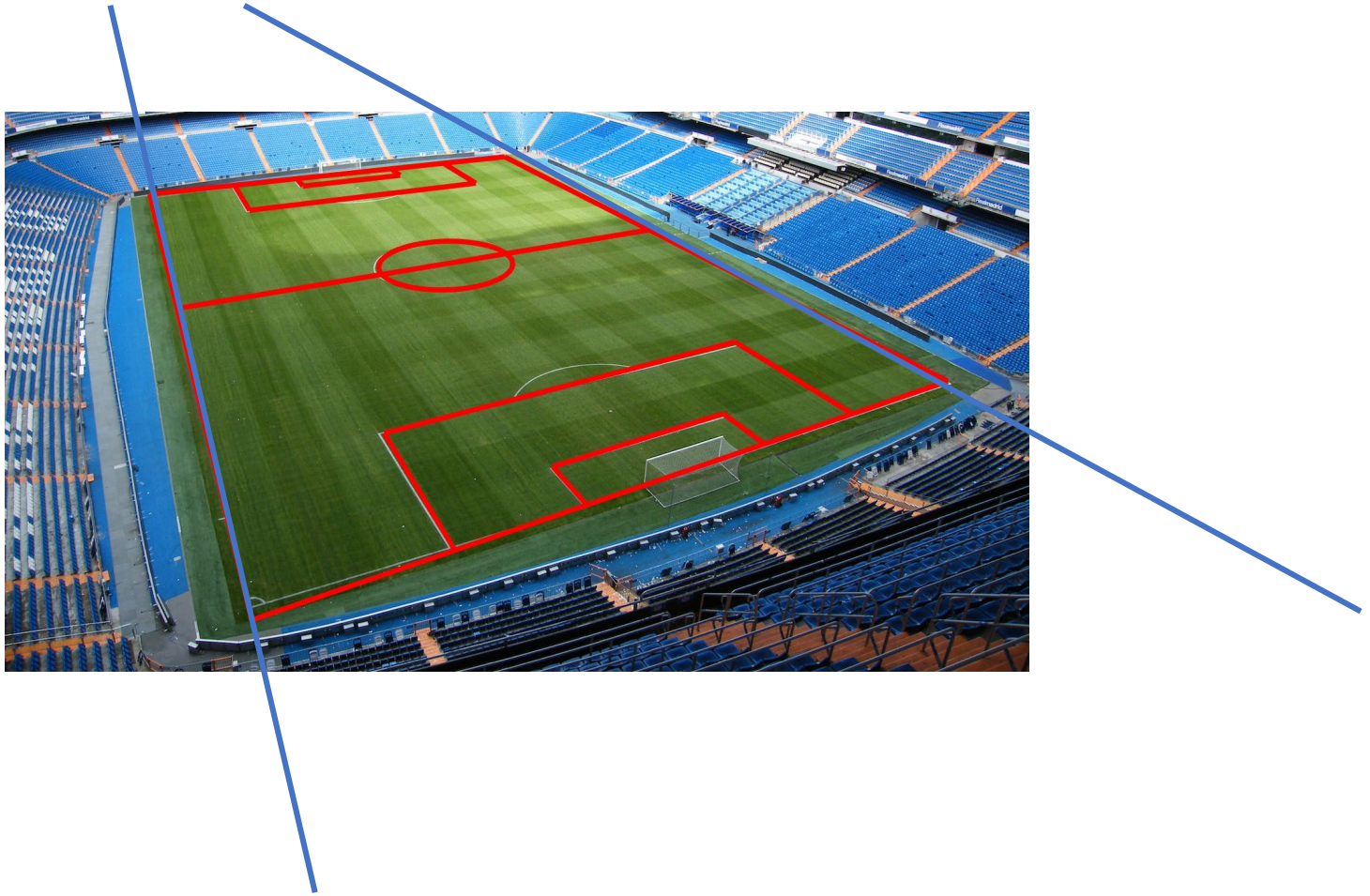


Bad news:
what you see IS NOT what you get

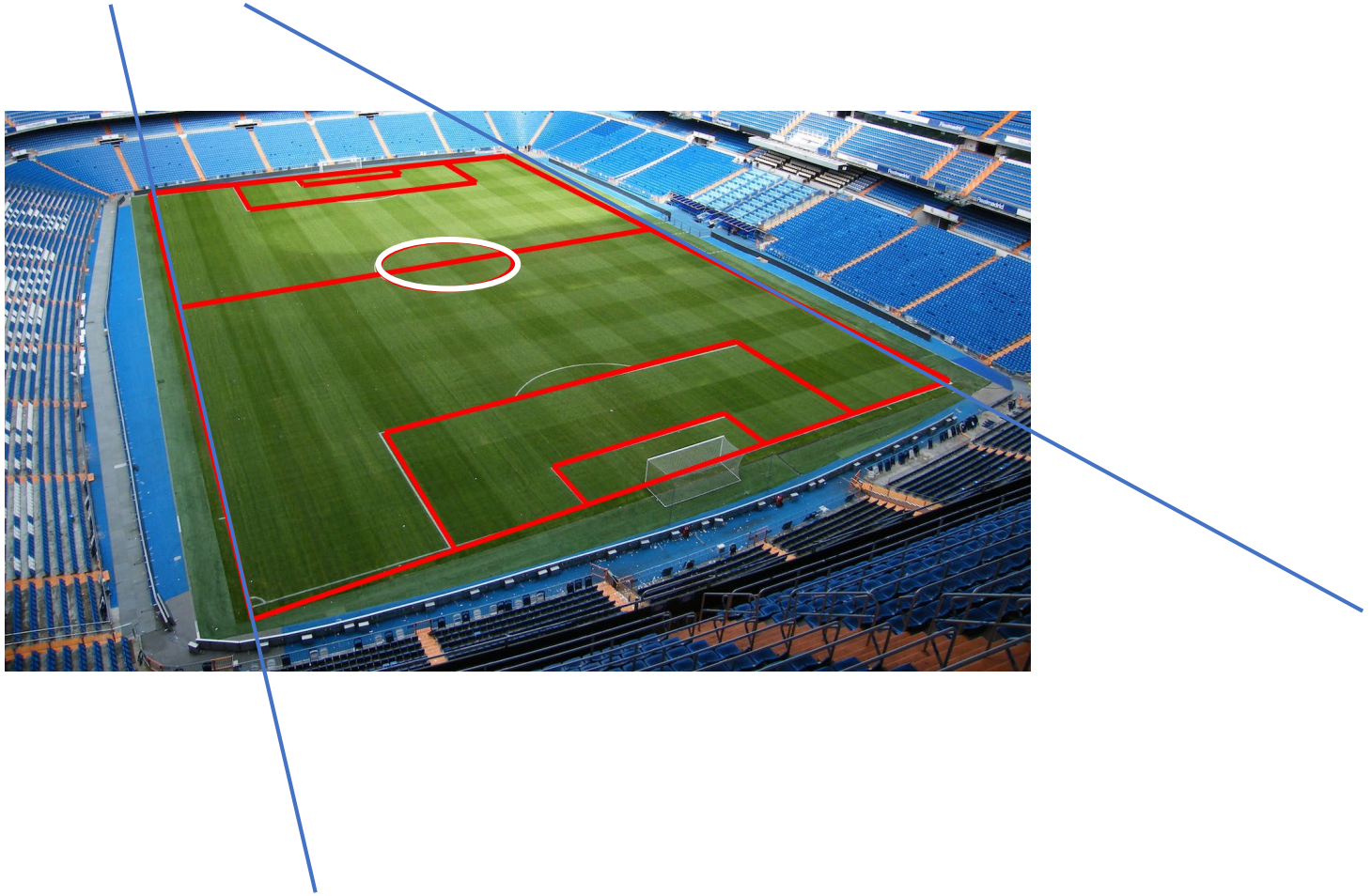
what you see **IS NOT** what you get



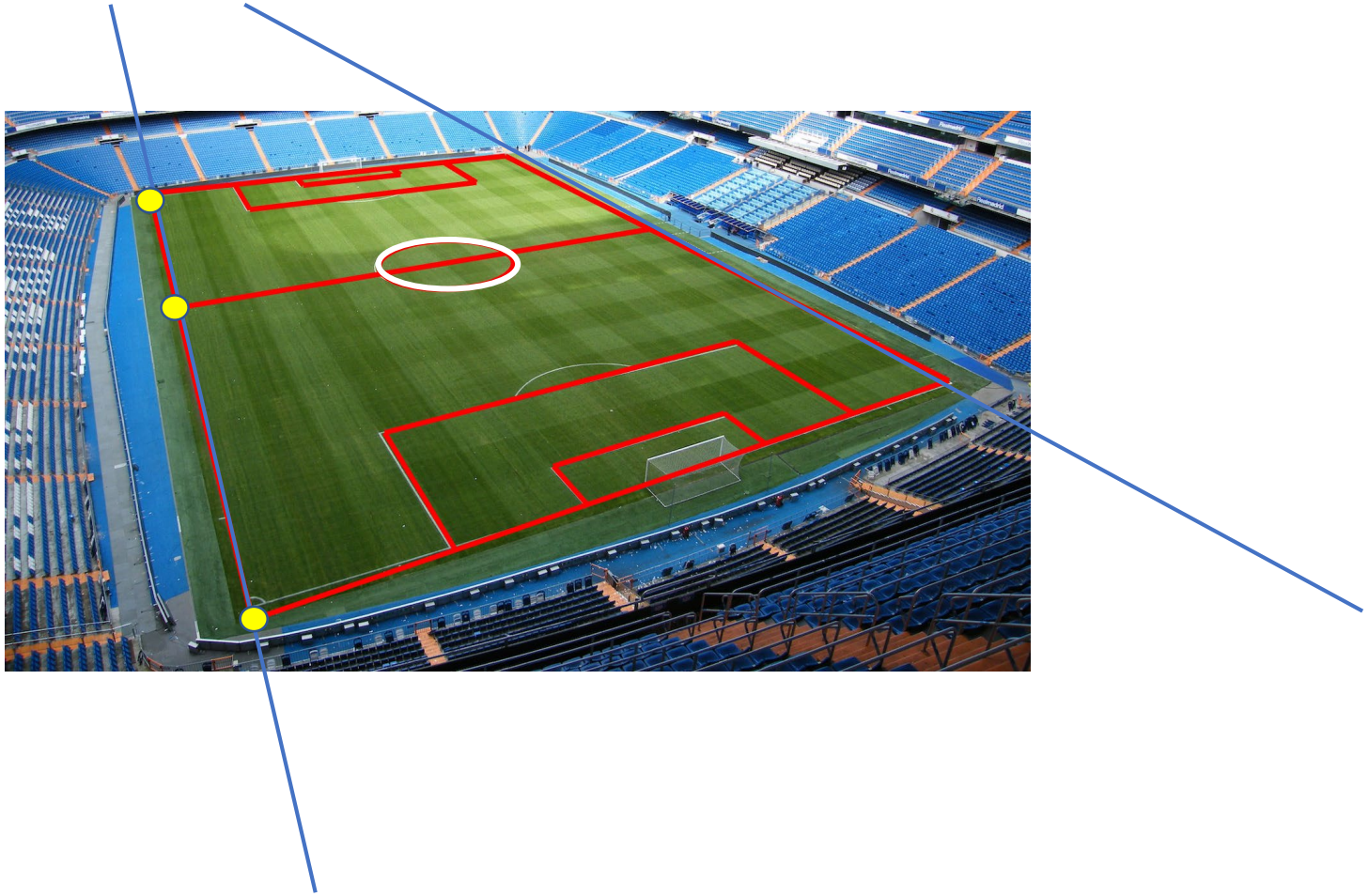
Parallel lines **DO NOT PROJECT ONTO** parallel lines



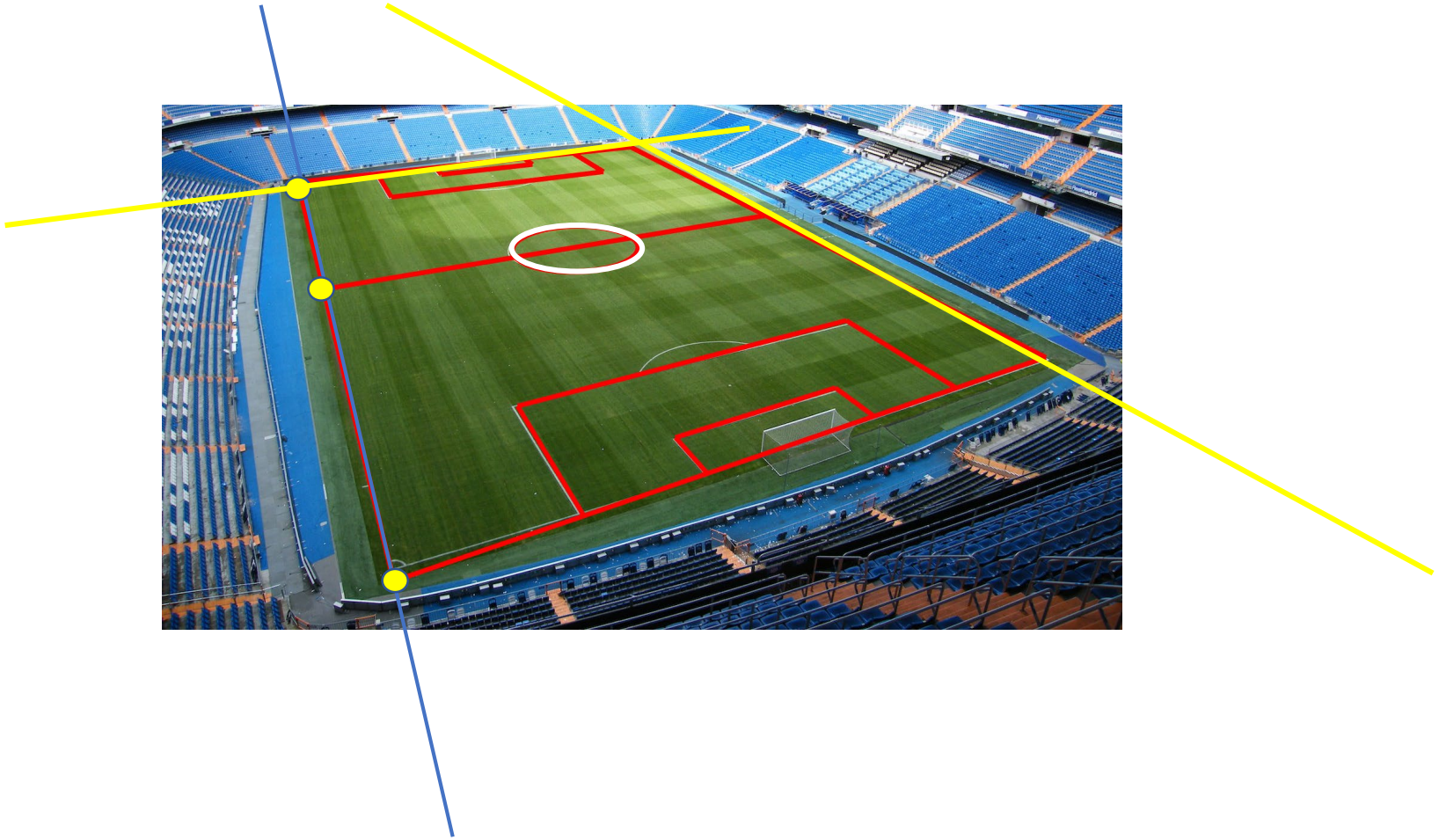
circumpherences **DO NOT PROJECT ONTO** circumferences



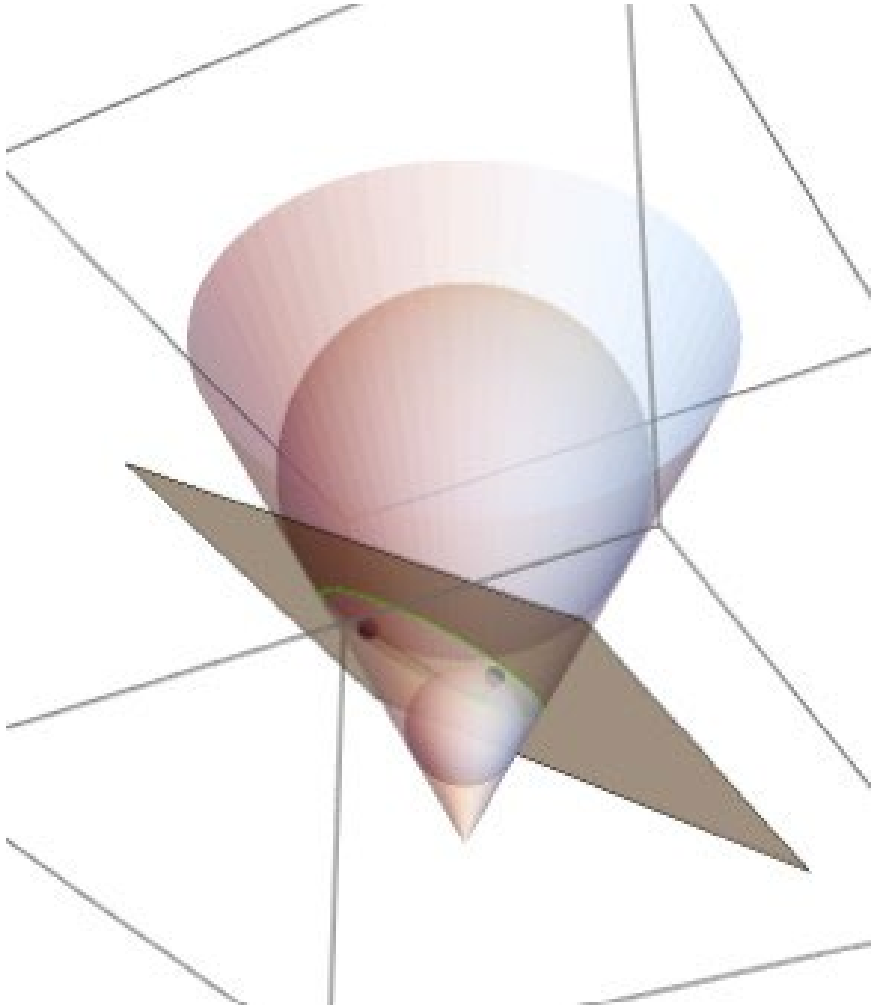
midpoints **DO NOT PROJECT ONTO** midpoints



Orthogonal lines **DO NOT PROJECT ONTO** orthogonal lines



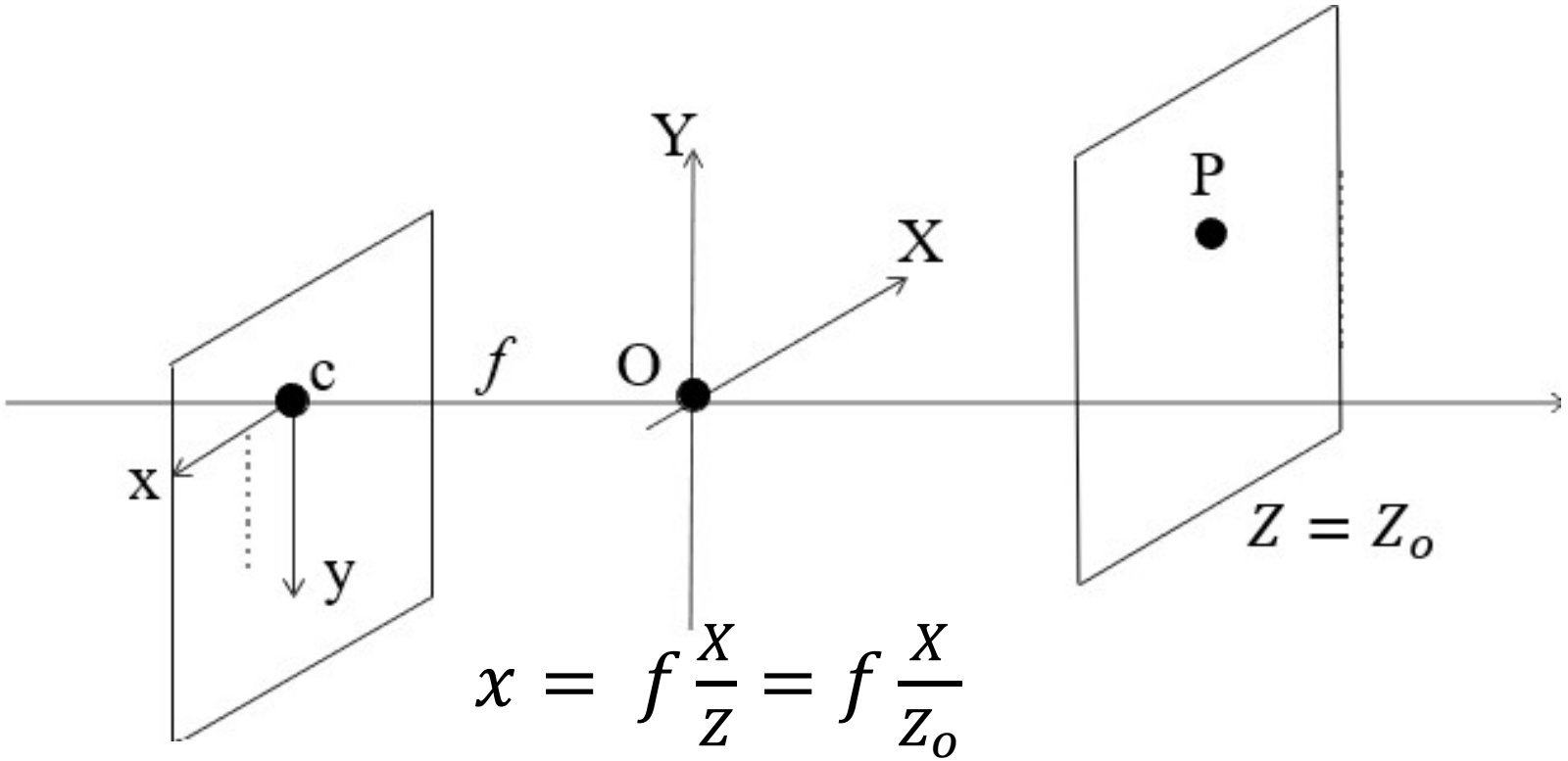
Sphere outlines **DO NOT PROJECT ONTO** circumpherences



Exception:

planar scene **parallel** to image plane

$$Z = Z_o = constant$$

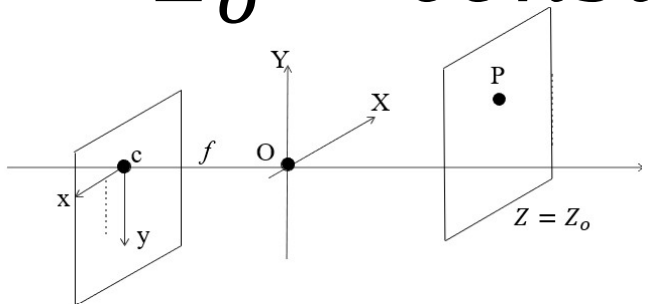


$$x = f\frac{X}{Z} = f\frac{X}{Z_o}$$

$$\rightarrow x = \text{const}\, X \longrightarrow$$ shape preserved

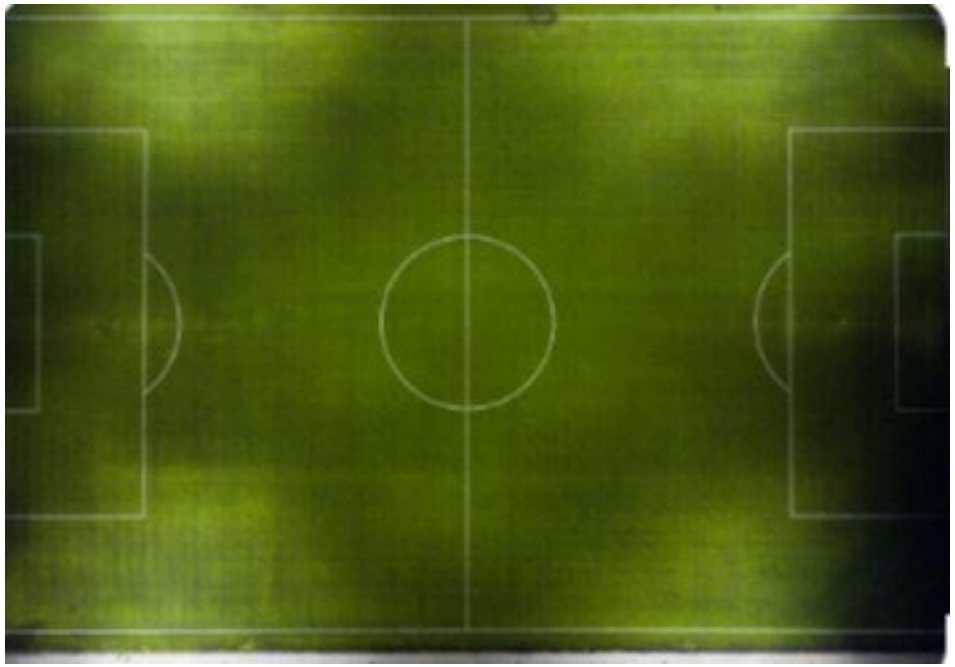
Exception:
planar scene **parallel** to image plane

$$Z = Z_o = constant$$



$$x = f\frac{X}{Z_o} = kX$$

$$y = f\frac{Y}{Z_o} = kY$$



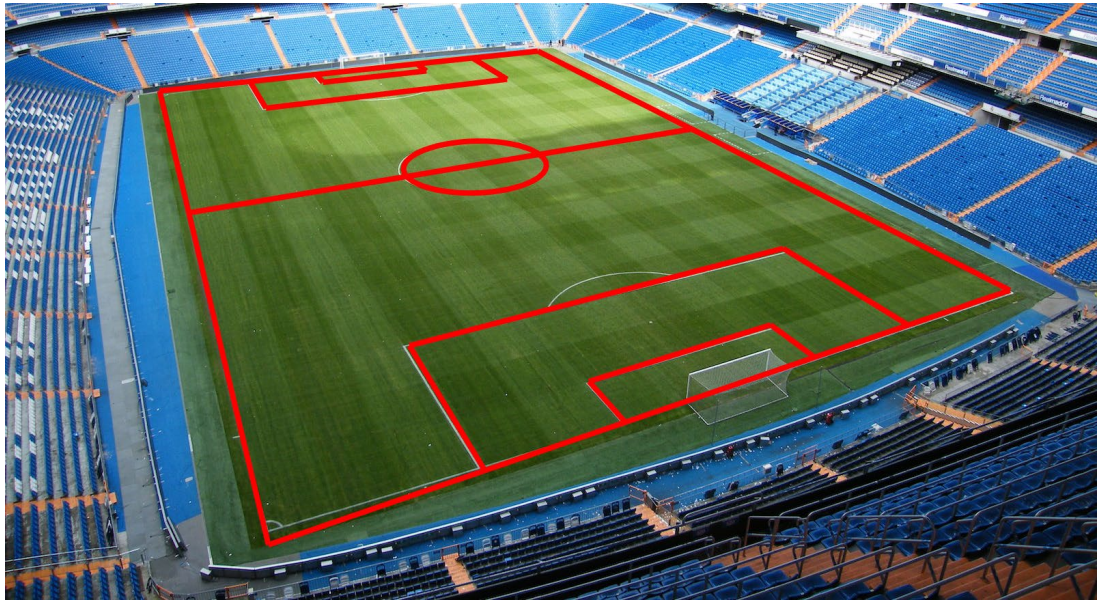
The image is a **scaled** version of the scene:

→ **same shape**

↓

angles preserved

what you see **IS NOT** what you get



if we want to estimate the SHAPE of what we are observing, we need to understand the
Geometry of the Projection